

# EasySave v3.0

## *Technical Specifications*

ProSoft — v3.0 — 2026

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## 1. Context

Following the deployment of EasySave v2.0 at multiple client sites, ProSoft management has identified a set of improvements required to meet growing client demands. This document defines the technical specifications for EasySave v3.0.

## 2. Objectives

- Replace sequential backup execution with full parallel execution
- Allow users to control each job in real time (Pause, Play, Stop)
- Implement priority file management across parallel jobs
- Prevent network saturation by limiting simultaneous large file transfers
- Enforce CryptoSoft mono-instance behaviour using a global Mutex
- Centralize daily log files across multiple machines via a Docker service

## 3. Functional Requirements

### 3.1 Parallel Backup

- All backup jobs must run simultaneously when Run All is triggered
- Each job must run in its own Task (Task.WhenAll)
- The UI must remain responsive during all backup operations
- Sequential mode is no longer required

### 3.2 Priority File Management

- User must be able to define a list of priority file extensions in Settings
- Files with priority extensions must be transferred before all other files
- This rule must apply across all parallel jobs simultaneously
- A non-priority file must not transfer if a priority file is pending in any job

### 3.3 Large File Transfer Limit

- User must be able to configure a maximum parallel file size in Ko
- Only one file exceeding the threshold may transfer at a time
- Files below the threshold may transfer freely in parallel
- Default threshold: 15,360 Ko (15 MB)

### 3.4 Pause / Play / Stop

- Each job must expose Pause, Play, and Stop controls in the UI
- Pause: backup pauses after the current file completes
- Play: backup resumes from where it stopped
- Stop: backup stops immediately — current file is abandoned
- Real-time progression (percentage) must be displayed per job

### 3.5 Business Software Detection (updated from v2.0)

- If business software is detected during backup: all jobs pause automatically

- When the software closes: all jobs resume automatically
- v2.0 behaviour (stop on detection) is replaced by this pause/resume logic
- Monitoring thread runs every 1000ms

### 3.6 CryptoSoft Mono-Instance

- CryptoSoft.exe must be modified to reject a second instance via global Mutex
- CryptoSoftService must wait for the Mutex before launching encryption
- Mutex name: Global\ CryptoSoftMutex
- If CryptoSoft is already running: wait and retry, do not skip encryption

### 3.7 Docker — Centralized Log Server

- A log server must be developed as an ASP.NET Core minimal API
- The server must be containerized with Docker and expose port 5000
- EasySave must send each log entry via HTTP POST to the server
- 3 log destination modes must be available: Local / Remote / Both
- The server must maintain one daily log file regardless of machine count
- Endpoints required: POST /log (receive entry), GET /health (status), GET /logs (read today's log)

## 4. Technical Requirements

| Requirement     | Specification                                                 |
|-----------------|---------------------------------------------------------------|
| Language        | C# / .NET 8.0                                                 |
| UI Framework    | WPF (Windows Presentation Foundation)                         |
| Architecture    | MVVM + Strategy + Singleton + Observer                        |
| Parallelism     | Task, async/await, Task.WhenAll                               |
| Synchronization | SemaphoreSlim, Mutex, ManualResetEventSlim, CancellationToken |
| Log library     | EasyLog.dll (separate project)                                |
| Encryption      | CryptoSoft.exe (mono-instance, XOR algorithm)                 |
| Docker          | ASP.NET Core minimal API, port 5000                           |
| Log format      | JSON or XML — configurable in Settings                        |
| File locations  | No c:\temp\ paths — use %AppData%                             |
| Code language   | English (comments and code)                                   |

## 5. Deliverables

| Deliverable         | Description                                |
|---------------------|--------------------------------------------|
| EasySave.exe v3.0   | WPF application with all v3.0 features     |
| EasyLog.dll v3.0    | Updated library with remote log support    |
| CryptoSoft.exe v3.0 | Updated with mono-instance Mutex           |
| LogServer (Docker)  | Containerized log centralization service   |
| User Manual v3.0    | Updated bilingual documentation            |
| Release Note v3.0   | Description of changes                     |
| UML Diagrams        | Updated class, sequence, use case diagrams |
| GitHub repository   | Tagged v3.0 release                        |

